

Enhancing open-circuit voltage in lead-free CsSnCl_3 perovskite solar cells via Rb incorporation: a combined DFT and SCAPS simulation study

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Abstract

This study thoroughly examines the properties and physical behavior of $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ by employing density functional theory as implemented in the Vienna Ab initio Simulation Package (VASP) software package. Our results demonstrate a notable enhancement in these properties due to Rb incorporation in CsSnCl_3 , enhancing the material's stability when serving as the photon-capturing layer in solar devices. $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ demonstrates robust structural stability, improved elastic characteristics and a good absorption coefficient. Thermal analysis indicates an affordable Debye and melting temperature, ensuring performance stability. To evaluate its performance in a solar cell, we examined the impact of various layer thicknesses (absorber, electron transport layer (ETL) and hole transport layer (HTL)), defect density, interfacial properties and shunt and series resistances on the device's output parameters using solar cell capacitance simulator (SCAPS) software. Our findings show that an optimized configuration of $\text{CuSbS}_2/\text{Cs}_{0.875}\text{Rb}_{0.125}\text{SnCl}_3/\text{WS}_2/\text{FTO}$ achieved an open-circuit voltage (V_{oc}) of 1.36 V, a short-circuit current density (J_{sc}) of 15.27 mA cm^{-2} , a fill factor (FF) of 88.26%, and a power conversion efficiency (PCE) of 18.39%. These outcomes underscore the capabilities of $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ as an efficient absorber substance for advanced photovoltaic devices, this led to improved voltage in tin-based solar cells.

Keywords: halide perovskite CsSnCl_3 and $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$, absorber material, solar cell, DFT calculations, SCAPS simulation, undefined

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1. Introduction

Since 2009, the perovskite solar cell (PSC) has proven its value in photovoltaic (PV) solar cells with an efficiency of 3.8% [1]. Recently, metal-halide perovskites (MHPs) have emerged as the most promising PV solar cells [2], with an efficiency over 26.1% for a single junction [3]. This is due to their interesting characteristics [4–6], which include the ability to adjust the band gap [7, 8], a high charge carrier mobility [9], the dispersal of charges over long distances [10], the capacity to absorb a significant amount of visible light [11], low reflectivity [12], low recombination rate [13, 14] and high ductility [15]. These perovskites are easy to fabricate and are widely available [16]. Therefore, they perform better in terms of efficiency and cost-effectiveness than the silicon-based PV systems [17]. There has been a growing interest in all-inorganic MHPs in their ABX_3 stoichiometry, particularly cesium (Cs)-based perovskite, shown as $CsBX_3$. This is mainly due to their remarkable thermal stability above 100 °C in contrast to methylammonium (MA) and formamidinium (FA), which has captured the attention of many researchers [18]. The main concerns are the harmful effects of lead (Pb) in the B-site and the instability of organic–inorganic halide perovskites under high-humidity environments. To address these concerns, there is a push to develop all-inorganic perovskites free of Pb content. Sn(II) and Ge(II), which belong to the same group in the periodic table as Pb(II), are regarded as primary candidates to replace Pb thanks to their comparable electronic characteristics. However, the current performance of Sn- and Ge-based all-inorganic perovskite-based solar cells is limited. For example, $CsGeI_3$ has an efficiency of only 0.2% [19], while $CsSnI_3$ has an efficiency of 3.31% [20], far from the desired values. $CsSnCl_3$, a Pb-free alternative, has gained interest due to its excellent stability and widespread availability for the industrial market. Unfortunately, $CsSnCl_3$'s optoelectronic performance is limited by its comparatively high band gap of 2.9 eV and its high absorption in the ultraviolet spectrum compared to the visible spectrum. The occupation of the A-site by a mixture of organic–inorganic cations demonstrates better absorption of sunlight in the visible spectrum and increased stability, as indicated by the Goldschmidt tolerance factors for ABX_3 compounds. It is important to mention that the atomic radius of Cs^+ or Rb^+ cations is smaller than that of MA^+ or FA^+ cations [21]. Thin films of $CsSnCl_3$ can be prepared by spin-coating precursor solutions of CsCl (or CsX) and anhydrous $SnCl_2$ in a DMF/DMSO solvent under an inert nitrogen atmosphere to prevent the oxidation of Sn^{2+} . The solution, with a concentration ranging from 0.3 to 0.8 M, is filtered and deposited using solvent-engineering techniques or an antisolvent drip method. It is then annealed at temperatures between 80 and 150 °C to control crystallization. To stabilize Sn^{2+} , additives such as excess $SnCl_2$, SnF_2 , or reducing agents are incorporated, while minimizing exposure to air and moisture [16].

Thanks to advancements in computational capabilities and approaches, the optical, mechanical and band gap properties

of halide perovskites are calculated using a first-principles method, which is focused on density functional theory (DFT) [22–24]. This approach, which has its roots in quantum physics, offers a fundamental basis for understanding the electronic configuration and characteristics of various materials. Through sophisticated computational simulations, the DFT approach enables a detailed analysis of how organic–inorganic halide perovskites interact with light. This offers significant insights for the evolution of perovskite-based technologies in PV cells. Numerous research teams have made substantial use of this method.

Several studies have investigated the behavior of inorganic Pb-free perovskites for PV applications [20, 25–28]. Siddique and Rahman [29] carried out research on using cubic $KGeCl_3$ as a light-harvesting layer in PV device structure. The study utilized DFT followed by a solar cell capacitance simulator (SCAPS) which revealed a remarkable efficiency of 15.83%. The research reported by Pindolia *et al* [27] utilized both DFT and SCAPS for the study. They focused on using orthorhombic $KSnI_3$ as a light-absorbing layer and attained an efficiency of 9.77%. Zhu *et al* [26] introduced Sb^{3+} into the crystal lattices of $CsPbI_2Br$ perovskite. They utilized a screen-printable nanocarbon-based electrode to substitute organic hole-transport materials and costly noble metal counter-electrodes. By incorporating 0.25% Sb, they achieved an efficiency of 11.06%. Park *et al* [30] investigated the impact of doping the A-site in $MAPbI_3$ with Rb by fabricating $Rb_{0.05}MA_{0.95}PbI_3$ with a Rb content of $x = 0.05$. This study demonstrated the stability and efficiency of the material, which achieved a power conversion efficiency (PCE) of 18.8%. Although various research teams have previously investigated the structural, mechanical, optical and electrical characteristics of $CsSnCl_3$ perovskite [31–33], no studies have been found in the literature regarding the incorporation of Rb into the A-site of $CsSnCl_3$ and its analysis as an absorber material in solar cell applications. The existence of this research gap has motivated us to conduct this study. Various semiconducting materials have been used in PSCs as the hole transport layer (HTL). Commonly employed HTL materials in PSC designs include Cu_2O [34], $CuSCN$ [35], $CuSbS_2$ [36], V_2O_5 [37], NiO [38] and MoO_3 [39], among others. Ideally, the HTL should exhibit high hole mobility, strong chemical stability, a direct band gap, low electron affinity and a high workfunction. Conversely, the electron transport layer (ETL) in PSC designs has utilized materials such as IGZO [40], SnO_2 [41], WS_2 [42], CeO [43], TiO_2 [44] and ZnO [45]. The ETL should possess strong transparency, a wide band gap, high electron mobility and appropriate alignment with the light-absorbing layer.

In this work, we have used a pseudo-potential approach implemented in VASP code and SCAPS software to numerically study the structural, mechanical, thermal, electronic, optical and performance characteristics of $CsSnCl_3$ and $Cs_7Rb_1Sn_8Cl_{24}$ alloys. $Cs_7Rb_1Sn_8Cl_{24}$ is sandwiched between the ETL and HTL layers to form a PSC. We have optimized the structure using different ETL and HTL materials to find the best candidate element. Our research proves that

incorporating Rb into the A-site of the perovskite CsSnCl_3 , used as an absorber material, improves the external parameters of the cells, which dramatically enhances the voltage in pure and sustainable solar cells based on Sn.

2. Computational methods

In this research, we explore the performance of a perovskite-based photovoltaic cell centered CsSnCl_3 with Rb doped in Cs-site with $x \sim 12\%$. Initially, we used the DFT method to examine the optical, electronic, structural, mechanical, and thermal behavior of CsSnCl_3 and $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$. Secondly, we used SCAPS software to analyze the perovskite embedded between the ETL and HTL layers to form a single-junction solar cell. We optimized their main parameters including the thickness of the absorber layer, ETL and HTL, defect density, interfacial properties, and shunt and series resistances. Finally, we investigated the cell's PCE, fill factor (FF), voltage and current density.

2.1. DFT calculations

The pseudo-potential method was performed within the DFT framework using the VASP code to compute the electronic structure of Cs-based perovskites [46, 47]. The interactions between ions and electrons are modeled using the projector-augmented wave method [48, 49]. The configuration electronic structure used for Cs, Sn and Cl is $[\text{Xe}]6s^1$, $[\text{Kr}]4d^{10}5s^25p^2$ and $[\text{Ne}]3s^23p^5$, respectively. The exchange-correlation energy is estimated using the generalized gradient approximation (GGA) [50] parameterized by Perdew–Burke–Ernzerhof (PBE) [51] and the GGA + U correction [4–6], which effectively improves calculations of electronic properties, especially energy gaps [7]. The additional GGA + U (U: Hubbard) correction [52] is made to analyze the impact of on-site Coulomb and exchange interactions on the optical and electronic characteristics of CsSnCl_3 perovskite and to compare the results with experimental data. The calculation of GGA + U is conducted using various U_{eff} values, ranging from 1 to 9 eV according to the Sn's s and p orbitals. The Brillouin zone (BZ) is sampled utilizing the Monkhorst-Pack technique [28] with a $15 \times 15 \times 15$ k -point grid. For the cubic structure of CsSnCl_3 , a plane wave energy cutoff of 650 eV is used.

Furthermore, the Cs atom is substituted with Rb in the crystal CsSnCl_3 at specific concentrations $x \sim 12.5\%$ in a $2 \times 2 \times 2$ supercell. These doping concentrations were selected with careful consideration of scientific and computational factors. Specifically, the 12.5% doping is the lowest practical concentration that can be achieved, given the limitations of our computational capacities. The electronic configuration of Rb is defined by $[\text{Kr}] 5s^1$. The atomic numbers for the cubic supercell of the perovskite are 7 (Cs), 1 (Rb), 8 (Sn), and 24 (Cl).

The mechanical features are determined using the Voigt–Reuss–Hill averaging method [53] based on the computed amounts of the elastic constants C_{ij} [54],

using the following equations, (1)–(3):

$$B_V = B_R = \frac{C_{11} + 2C_{12}}{3} \quad (1)$$

$$G_V = \frac{C_{11} - C_{12} + 3C_{44}}{5} \quad (2)$$

$$G_R = \frac{5C_{44}(C_{11} + 2C_{12})}{4C_{44} + 3(C_{11} - C_{12})}. \quad (3)$$

Employing the values of G_R , G_V , B_R and B_V , when R refer to Reuss and V refer to Voigt, the average mathematical values of the bulk (B) and shear modulus (G) were calculated using Hill's approximation [55], equations (4) and (5):

$$B = \frac{B_V + B_R}{2} \quad (4)$$

$$G = \frac{G_V + G_R}{2}. \quad (5)$$

The following formulas (6)–(8) were used to obtain Young's modulus (E) and Poisson's ratio (ν) employing B , G and the anisotropy factor (A) [55]:

$$E = \frac{9BG}{(3B + G)} \quad (6)$$

$$\nu = \frac{3B - 2G}{2(3B + G)} \quad (7)$$

$$A = \frac{2C_{44}}{C_{11} - C_{12}}. \quad (8)$$

Debye temperature and melting point are both relevant in comprehending the thermal characteristics of materials in solid-state physics. Their interaction is especially significant in designing and manufacturing nanomaterials for PVs, where heat management significantly influences device efficiency and life span.

The Debye temperature represents a distinguishing temperature that offers insight into the phonon spectrum of a solid. It is defined based on the Debye model, which approximates the vibrational modes of a crystal lattice. This temperature is significant because it helps estimate the heat capacity of solids at low temperatures. Specifically, it indicates the temperature below which the heat capacity follows the Debye model, typically exhibiting a T^3 dependence. Higher Debye temperatures generally suggest stronger bonding and greater stiffness in materials [56]. The melting point is the temperature at which a solid transitions to a liquid state. At this temperature, the solid and liquid states co-exist in balance. The melting point varies with pressure and is typically specified at standard atmospheric pressure (1 atm) [57].

The Debye temperature can be calculated based on elastic constants. In solid materials, it can be approximated using the

average sound velocity (ν_m), longitudinal acoustic wave velocity (ν_l) and transverse acoustic wave velocity (ν_s) through the following equations (9)–(12):

$$\theta_D = \frac{h}{k} \left[\frac{3n}{4\pi} \left(\frac{N_A \rho}{M} \right) \right]^{1/3} \nu_m \quad (9)$$

$$\nu_m = \left[\frac{1}{3} \left(\frac{2}{\nu_s^3} + \frac{2}{\nu_l^2} \right) \right]^{-1/3} \quad (10)$$

$$\nu_s = \left(\frac{3B + 4G}{3\rho} \right)^{1/2} \quad (11)$$

$$\nu_l = \left(\frac{G}{\rho} \right)^{1/2} \quad (12)$$

where N_A represents Avogadro's number, k represents Boltzmann's constant, h represents Plank's constant, ρ is the density and M is the molecular weight for each formula unit.

The melting temperature, T_m , of perovskites can also be determined using equation (13) [56]:

$$T_m = 553 + \frac{(5.91 \times C_{11})}{G}. \quad (13)$$

To investigate the optical properties, the real and imaginary components of the frequency-dependent dielectric functions, $\varepsilon(\omega) = \varepsilon_1(\omega) + i\varepsilon_2(\omega)$, have been calculated utilizing DFT [57]. The formulas listed below in equations (14)–(17) were employed to compute $\alpha(\omega)$ (absorption coefficient), $n(\omega)$ (refractive index), $L(\omega)$ (optical loss factor), and $R(\omega)$ (reflectivity) based on the values of $\varepsilon_1(\omega)$ and $\varepsilon_2(\omega)$.

$$n(\omega) = \frac{1}{\sqrt{2}} \left[\left[\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega) \right]^{1/2} + \varepsilon_1(\omega) \right]^{1/2} \quad (14)$$

$$\alpha(\omega) = \sqrt{2}\omega \left[\left[\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega) \right]^{1/2} - \varepsilon_1(\omega) \right]^{1/2} \quad (15)$$

$$R(\omega) = \left| \frac{\sqrt{\varepsilon(\omega)} - 1}{\sqrt{\varepsilon(\omega)} + 1} \right|^2 \quad (16)$$

$$L(\omega) = \frac{\varepsilon_2(\omega)}{\varepsilon_1^2(\omega) + \varepsilon_2^2(\omega)}. \quad (17)$$

Using the band structure data surrounding the Γ point of the BZ, the effective masses of holes (m_h^*) and electrons (m_e^*) can be calculated [58]. This is done by estimating the second-order derivatives of the conduction band (CB) and valence band (VB) edges with respect to the wave vector k [59]. The following formula, equation (18), can be used to obtain the effective masses:

$$m^* = \left(\frac{\hbar}{\frac{\partial^2 E}{\partial k^2}} \right). \quad (18)$$

The effective mass (m^*) of the electron and hole is used to calculate the effective density of states in the CB (N_c) and VB (N_v). Formula (19) for the effective density of states N_{eff} is as follows [60]

$$N_{\text{eff}=2g_d} \left(\frac{2\pi TK_b m^*}{h^2} \right)^{3/2} \quad (19)$$

where g_d, K_b, m^*, T and h are the factor of degeneracy, Boltzmann's constant, effective mass, temperature, and Planck's constant, respectively.

Using parameters derived from DFT, SCAPS software simulates the behavior of a PSC based on $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$.

2.2. SCAPS-1D numerical simulation of solar cell device

This paper reveals the vital aspects that significantly affect the performance of the device. Specifically, it explores how semiconductor materials behave under stable conditions and how this behavior can be numerically solved with SCAPS-1D software [61]. The software provides a numerical model to find solutions of the one-dimensional Poisson and carrier continuity equations to get an output PCE, FF, short-circuit current density (J_{sc}), current–voltage characteristics and open-circuit voltage (V_{oc}) [62].

Sunlight is assumed to be AM 1.5 G (1000 Wm^{-2}) in all simulations. When radiative recombination between free electrons and holes in PSCs is minimal, the default models for radiative band-to-band and Auger recombination are ignored [35]. The emphasis of this study is on recombination caused by defects. The PSC is constructed and evaluated utilizing the structure (FTO/ETL/ $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ /HTL). Bulk defect densities are employed as the control variables to maintain consistency and comparability with the findings. To simplify the analysis, only a single bulk defect level is examined for each material.

The following are the Poisson's equation and continuity equations (20)–(22) for holes and electrons, respectively, which were employed in the numerical simulation:

$$-\frac{\partial}{\partial x} \left(-\varepsilon(x) \frac{\partial V}{\partial x} \right) = q [p(x) + p_t(x) - n(x) - n_t(x) + N_D^+(x) - N_A^-(x)] \quad (20)$$

$$\frac{\partial p}{\partial t} = \frac{1}{q} \frac{\partial J_p}{\partial x} + G_p - R_p \quad (21)$$

$$\frac{\partial n}{\partial t} = \frac{1}{q} \frac{\partial J_n}{\partial x} + G_n - R_n \quad (22)$$

where $n(x)$ and $p(x)$ represent the concentration of free electrons and holes, respectively. V , q and ε are the potential, charge and permittivity, respectively. R and G stand for the recombination and generation rates, respectively, and N_D^+ and N_A^- are the concentrations of ionized donor and acceptor, respectively. The concentrations of holes and electrons in

Table 1. Input parameters of hole transport layers (HTLs).

HTL	Cu ₂ O	CuSbS ₂	CuO	NiO	CuI
Thickness (μm)	0.15	0.15	0.15	0.15	0.15
E _g (eV)	2.2	1.58	1.51	3.8	3.1
χ (eV)	3.4	4.2	4.07	1.46	2.1
ϵ_r	7.5	14.6	18.1	10.7	6.5
N_c (cm ⁻³)	2×10^{19}	2×10^{18}	2.2×10^{19}	2.8×10^{19}	2.8×10^{19}
N_v (cm ⁻³)	10^{19}	2×10^{19}	5.5×10^{20}	10^{19}	10^{19}
μ_n (cm ² /VS)	200	49	100	12	100
μ_p (cm ² /VS)	8600	49	0.1	2.8	43.9
N_A (cm ⁻³)	10^{18}	10^{18}	10^{18}	10^{18}	10^{18}
N_D (cm ⁻³)	0	0	0	0	0
N_t (cm ⁻³)	10^{14}	10^{14}	10^{14}	10^{14}	10^{14}
References	[22, 63]	[63]	[64]	[65]	[66]

Table 2. Input parameters of the electron transport layers (ETLs).

ETL	TiO ₂	ZnO	WS ₂	SnS ₂
Thickness (μm)	0.145	0.145	0.145	0.145
E _g (eV)	3.2	3.3	1.8	2.24
χ (eV)	4	4	3.95	4.24
ϵ_r	9	9	13.6	10
N_c (cm ⁻³)	2.10^{18}	$3.7.10^{18}$	1.10^{18}	$2.2.10^{18}$
N_v (cm ⁻³)	$1.8.10^{18}$	$1.8.10^{19}$	$2.4.10^{19}$	$1.8.10^{19}$
μ_n (cm ² /VS)	20	100	100	500
μ_p (cm ² /VS)	10	25	100	500
N_A (cm ⁻³)	0	0	0	0
N_D (cm ⁻³)	1.10^{21}	1.10^{18}	1.10^{18}	1.10^{17}
N_t (cm ⁻³)	1.10^{14}	1.10^{14}	1.10^{14}	1.10^{14}
References	[63, 67]	[63, 68]	[42]	[69]

their respective trap states are denoted by p_t and n_t , respectively. Calculation of the cell current and voltage is given by equation (23):

$$V_{oc} = \frac{KT}{q} \ln \left[\frac{J_{sc}}{J_0} + 1 \right] \quad (23)$$

where T is the temperature and q the charge of an electron. The variable J_0 stands for electron charge reverse saturation current density, while J_{sc} is the short-circuit current density and K is Boltzmann's constant.

In the last step, we compute the perovskite cell with various HTL and ETL candidate materials (tables 1 and 2) to optimize the device with the most sustainable and most efficient energy consumption.

3. Results and discussion

3.1. Structural and elastic properties

The cubic arrangement of the lead-free perovskite CsSnCl₃ is defined by space group Pm3m (No. 221) [70]. This compound has the following atomic arrangements (figure 1): according to Wyckoff position 1a, Cs atoms are located at the corner position (0, 0, 0), Sn atoms are situated in the position of the body center (0.5, 0.5, 0.5) and Cl atoms are situated at the face center position (0, 0.5, 0.5) [70]. Figure 1(a) displays the unit cell

of CsSnCl₃. In cubic CsSnCl₃ perovskite, the atoms arrange into a perfect crystal lattice where [SnCl₆] units connect at their corners to form an octahedral framework. In the heart of a cube of eight octahedral atoms, the Cs cation is situated in a 12-fold coordination site [71].

To incorporate doping into CsSnCl₃ perovskite, we utilized the supercell technique through the construction of a $2 \times 2 \times 2$ supercell composed of 40 atoms distributed as follows: 24 Cl, 8 Cs and 8 Sn. In order to achieve a doping concentration of 12.5%, we substituted one of the Cs atoms of the Cs₈Sn₈Cl₂₄ alloy with dopant rubidium (Rb) atoms. The Rb atom occupies the site (0.5, 0.5, 0.5) in the $2 \times 2 \times 2$ supercell. An illustration of the compound is presented in figure 1(b). Furthermore, the energy of formation (E_{form}) of CsSnCl₃ and Cs₇Rb₁Sn₈Cl₂₄ can be calculated using equation (24). In this equation, E^+ represents the total energy of the ABX₃ perovskite, while E_A , E_B and E_X represent the total energies of the A, B and X atoms, and the number of atoms of types A, B and X are x , y and z , respectively:

$$E_{form} = \frac{1}{x+y+z} [E_0 - (xE_A + yE_B + zE_X)]. \quad (24)$$

The formation energy of the cubic CsSnCl₃ is evaluated to be -1.458 eV, which is similar to the findings of Körbel *et al* [73] (-1.460) and Alqahtani *et al* [74] (-1.490). Additionally, the formation energy of cubic Cs₇Rb₁Sn₈Cl₂₄ is determined to

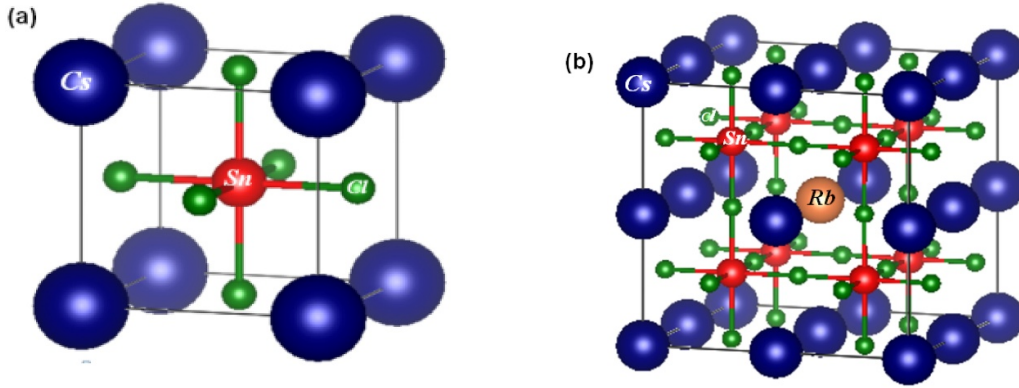


Figure 1. Cubic structure of the perovskite pure CsSnCl₃ (a) and (b) Rb-doped CsSnCl₃ visualized by VESTA software [72].

Table 3. Calculated lattice parameter (a), unit cell volume (V) and formation energy (E_f) of CsSnCl₃ and Cs₇Rb₁Sn₈Cl₂₄ perovskites.

Compound	Comparison with Others	a (Å)	V (Å ³)	E_f (eV)
CsSnCl ₃	This work	5.632	178.653	−1.458
	Experimental work	5.573 [80]	—	—
		5.583 [81]	174.02	—
	Other work	5.6458 [33]	179.883	—
		5.620 [31]	177.504	—
		5.4901 [78]	165.469	—
		5.610 [79]	—	—
		5.617 [32]	177.265	—
Cs ₇ Rb ₁ Sn ₈ Cl ₂₄	This work	5.626	178.121	−0.7483

be −0.7483 eV. The negative formation energy values suggest that the cubic phase of CsSnCl₃ and Cs₇Rb₁Sn₈Cl₂₄ are thermodynamically stable [75].

The Goldschmidt tolerance factor serves as a practical empirical indicator for predicting the preferred and stable structure of inorganic–organic hybrid halide perovskite materials. When the tolerance factor ranges between 0.8 and 1, where the octahedral factor is $\mu > 0.41$, these materials tend to form a stable cubic structure. To calculate the effective Goldschmidt tolerance factor (t_{eff}) of Cs₇Rb₁Sn₈Cl₂₄, we have used the effective ionic radii (r_{eff}) found in [71] in the following equations (25)–(27):

$$r_{\text{eff}} = x r_{\text{Rb}}^+ + (1 - x) r_{\text{Cs}}^+ \quad (25)$$

$$t_{\text{eff}} = \frac{r_{\text{eff}} + r_{\text{Cl}^-}}{\sqrt{2}(r_{\text{Sn}^{2+}} + r_{\text{Cl}^-})} \quad (26)$$

$$\mu = \frac{r_{\text{Sn}^{2+}}}{r_{\text{Cl}^-}}, \quad (27)$$

where $r_{\text{Cs}^+} = 1.88 \text{ Å}$, $r_{\text{Sn}^{2+}} = 1.18 \text{ Å}$, $r_{\text{Cl}^-} = 1.81 \text{ Å}$, $r_{\text{Rb}^+} = 1.72 \text{ Å}$ [76]. The tolerance factor of CsSnCl₃ is 0.873, which matches with the findings of Khachia *et al* [33] (0.897) and Travis *et al* [77] (0.95). Additionally, the tolerance factor of Cs₇Rb₁Sn₈Cl₂₄ is determined to be 0.8679. With an octahedral factor of $\mu = 0.652$, both CsSnCl₃ and Cs₇Rb₁Sn₈Cl₂₄ perovskites exhibit a stable cubic structure.

The unit cell volumes and lattice parameters, computed for CsSnCl₃ and Cs₇Rb₁Sn₈Cl₂₄ and shown in table 3, were found comparable to values obtained in previous theoretical works [30–33, 78, 79], as well as in excellent agreement with values reported in previous experimental works [80, 81]. The results match significantly with the experimental work of Ali *et al* [81], in which hot-injection nanocrystals (colloidal CsSnCl₃ NCs) were utilized rather than the common experimental methods of solution processing for thin films (spin-coating/two-step) [16]. This illustrates the validity of our DFT-based investigation. The dopant element Rb⁺ (1.72 Å) has a smaller ionic radius than Cs⁺ (1.88 Å), resulting in lower estimated values for the volume and lattice of the unit cell of the doped perovskite.

The computed total energies as a function of volume for cubic CsSnCl₃ and Rb-doped CsSnCl₃ using GGA-PBE approximations are graphically plotted in figure 2. It is evident that the Rb-doped CsSnCl₃ compound exhibits a marginally lower energy than the pure CsSnCl₃ phase. The attributes of the ground state and static equilibrium characteristics are fitted using the Murnaghan equation of state [54].

The elastic properties of the crystal structures of CsSnCl₃ and Cs₇Rb₁Sn₈Cl₂₄ perovskites are determined by employing the GGA-PBE approach and applying the stress–strain relationship [54]. In the case of the cubic system, there exist three independent elastic constants, C_{11} , C_{12} and C_{44} . To evaluate these constants, we considered four data points within the deformation range [0–0.04] for both orthorhombic and tetragonal strain modes. We plotted the total energy variation

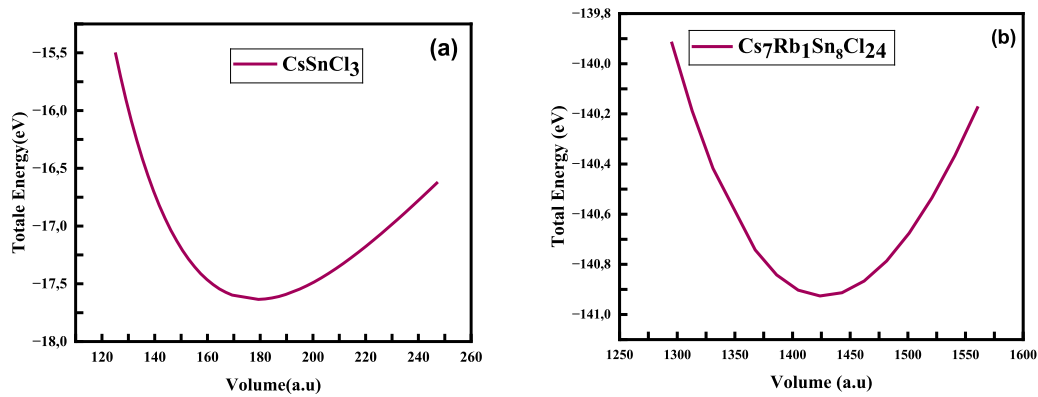


Figure 2. Total energy versus volume per unit cell of CsSnCl₃ (a) and Cs₇Rb₁Sn₈Cl₂₄(b) perovskites using GGA-PBE.

Table 4. Elastic constants C_{ij} of CsSnCl₃ and Cs₇Rb₁Sn₈Cl₂₄ (in GPa).

Material	C_{11}	C_{12}	C_{44}	$C_{12} - C_{44}$
CsSnCl ₃	50.207	9.503	5.900	3.603
Other work [32]	50.132	9.143	5.68	3.463
Other work [33]	44.075	12.322	5.603	6.72
Other work [70]	50.66	8.71	6.01	—
Cs ₇ Rb ₁ Sn ₈ Cl ₂₄	54.709	10.792	7.903	2.889

Table 5. Calculation of bulk modulus (B), shear modulus (G), Young's modulus (E), Poisson's ratio (ν), Vickers hardness (H_v) and Isotropic Poisson's Ratio (A) of pure CsSnCl₃ and Cs₇Rb₁Sn₈Cl₂₄ alloy compared with other theoretical studies.

Material	B (GPa)	G (GPa)	E (GPa)	ν	A	B/G	H_v (GPa)
CsSnCl ₃	23.071	9.961	26.123	0.311	0.311	2.316	1.289
Other work [33]	22.91	8.64	23.02	0.33	0.35	2.65	—
Other work [70]	22.70	10.20	26.61	0.300	—	2.22	—
Other work [32]	22.815	9.794	25.704	0.3122	—	2.329	1.226
Cs ₇ Rb ₁ Sn ₈ Cl ₂₄	25.431	12.074	31.273	0.295	0.30	2.106	2.301

as a function of the squared strain values [55]. The resulting elastic constants are summarized in table 4. The following equations (28)–(31) provide the stability conditions for the cubic phase according to Born [82]:

$$C_{11} - C_{12} > 0 \quad (28)$$

$$C_{11} + 2C_{12} > 0 \quad (29)$$

$$C_{44} > 0 \quad (30)$$

$$C_{12} < B < C_{11} \quad (31)$$

where B is bulk modulus. The elastic constants, computed as shown in table 4, meet the stability criteria, demonstrating that both CsSnCl₃ and Cs₇Rb₁Sn₈Cl₂₄ crystals are mechanically stable.

From table 4, we observe that substituting the Rb atom at the Cs-site in the Cs₈Sn₈Cl₂₄ alloy resulted in an increase in the elastic moduli (C_{ij} -Cs₇Rb₁Sn₈Cl₂₄ > C_{ij} -CsSnCl₃).

3.2. Mechanical properties

Using the previously determined elastic moduli C_{ij} , the mechanical properties of CsSnCl₃ and Cs₇Rb₁Sn₈Cl₂₄ were calculated. These properties include bulk modulus (B), shear modulus (G), B/G ratio, Young's modulus (E), and Poisson's ratio (ν), which play a crucial role in determining the strength of the material. The results of these calculations are presented in table 3. To obtain these values, the Voigt and Reuss limits for the bulk and shear moduli of the cubic crystal were calculated utilizing equations (1)–(6) and (8) [55].

As shown in table 5, there is a strong concordance between the mechanical properties of the perovskite in our research and those reported in previous theoretical investigations. One notable characteristic that demonstrates the resistance of the material to compression is its bulk modulus. Materials with a relatively high bulk modulus can resist greater compressive loads without deforming. An increased shear modulus denotes increased resistance to deformation under shear stress. Substances with a higher Young's modulus are generally more elastic and less prone to stretching or bending. The capability of a material to overcome deformation or damage within a load is quantified by its Vickers hardness (H_v). A higher Vickers

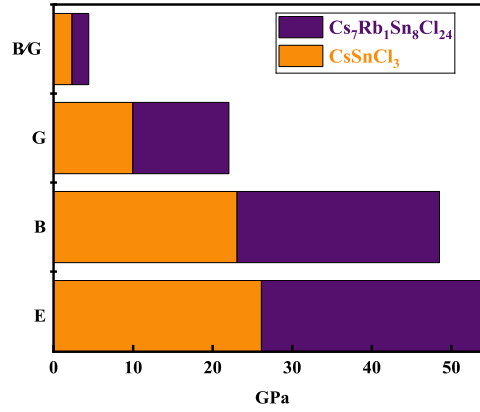


Figure 3. Bulk modulus (B), shear modulus (G), Young's modulus (E), and B/G factor of the pure and Rb-doped CsSnCl_3 .

Table 6. The output calculation of Debye and melting temperatures of CsSnCl_3 and $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$.

Material	ν_s	ν_l	ν_m	θ_D	$T_m \pm 300\text{K}$
CsSnCl_3	1726.140	3297.573	1930.841	174.808	282.79
Other work [33]	—	—	—	187.72	—
$\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$	1919.642	3560.170	2142.857	193.8	279.78

hardness implies that the material has a lower possibility of changing shape when subjected to a load.

Our study found that doped $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ with a cubic structure is more resistant than pure CsSnCl_3 . $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ showed considerably greater values of B , G and E , and an H_v of 2.301 GPa. This makes it an ideal material mechanically for optoelectronics applications, where a lower B is required in thin-film applications to enable the film to flex and adapt to the device's structure. When compared with CsSnCl_3 , both materials showed G and E values that are suitable for use in thin-film technology (see figure 3).

Furthermore, Poisson's ratio is used to identify the type of bonding forces inside a crystal, while together with Pugh's ratio, it is utilized for classifying materials either ductile or brittle [33]. The pivotal value of 0.26 serves as a boundary between brittle and ductile materials. The critical value for Pugh's ratio is 1.75, which acts as a separator between brittle and ductile materials. Values higher than this limit are considered ductile, while values lower than this are considered brittle. The results of our investigation indicate that CsSnCl_3 and $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ exhibit ionic bonding with Poisson's ratio of 0.311 and 0.295, respectively, as well as ductile behavior with a Pugh ratio of 2.316 and 2.106, respectively. Therefore, the Rb-doped phases are well suited for use in solar cell devices.

3.3. Thermal properties

The study focused on investigating the thermodynamic properties, typically the Debye temperature (θ_D) and melting temperature (T_m), which are vital in characterizing the thermal vibrations of atoms. This parameter is closely associated with various physical properties, including specific heat and melting

temperature. Furthermore, it is a valuable tool in assessing structural stability and bond strength.

The computed values of ν_m , ν_l and ν_s are presented in table 6. The acoustic velocities are greatest for the $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ alloy, suggesting that the propagation of acoustic waves in this structure is notably faster. In addition, the Debye temperature of cubic $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ is higher than that calculated for pure CsSnCl_3 . The melting temperature is a crucial factor in the field of engineering. For the cubic arrangement, the computed melting temperature (T_m) is determined to be $282.79 \pm 300\text{K}$ for pure CsSnCl_3 and $279.78 \pm 300\text{K}$ for $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ utilizing the GGA-PBE approximation.

3.4. Electronic properties

3.4.1. Band structure. The band structures of pure and Rb-doped CsSnCl_3 in a cubic structure are shown in figure 4. The Fermi level (EF) is indicated at zero on the energy scale, ranging from -3 eV to $+3\text{ eV}$. According to semiconductor theory, the properties of a solid substance are greatly influenced by the energy band structure near the EF. Consequently, we focused on examining the CB's lower section and of the VB's peak close to the EF. As shown in figure 4(a), CsSnCl_3 is a direct band gap semiconductor, with the lower part of the CB and the VB peak position existing at the gamma-point. Other symmetry sites (X–M–R) also display a direct band gap. Our GGA-PBE calculations indicate that the direct band gap of CsSnCl_3 is 0.9273 eV, confirming the accuracy of the work reported by Islam *et al* [31] and Roknuzzaman *et al* [70]. This demonstrates the resilience of our DFT studies. However, it is important to note that the simulated band gap obtained with GGA-PBE is lower than the value seen in experiments of 2.96 eV [81]. This general limitation of the GGA-PBE method appears due to correlation effects and

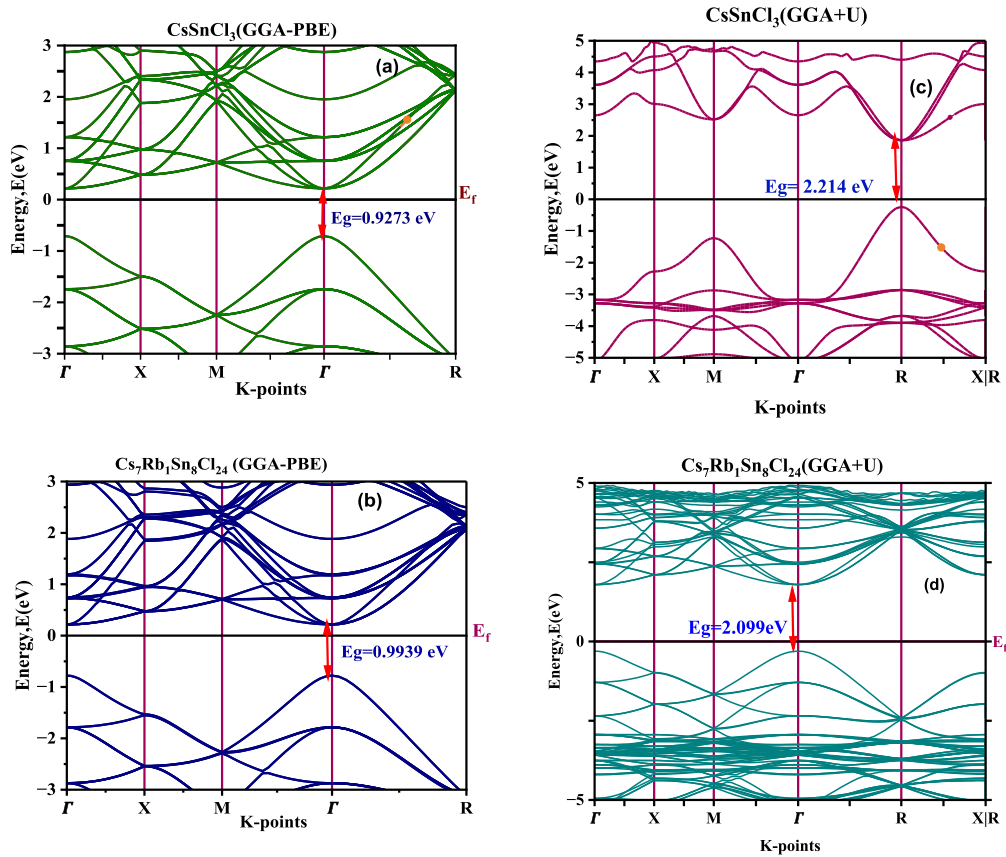
Table 7. Band gap and lattice characteristics of CsSnCl₃ for various U_{eff} values determined via GGA-PBE ($U_{\text{eff}} = 0$ eV) and GGA + U computations and the corresponding experimental values.

U_{eff} (eV)	0	1	2	3	4	5	6	7	8	9
E_g (eV)	0.9273	1.308	1.441	1.568	1.689	1.805	1.915	2.02	2.119	2.214
a (Å)	5.6321	5.6321	5.6334	5.6349	5.6352	5.6371	5.6344	5.6355	5.6364	5.6370
$\frac{a-a_{\text{exp}}}{a_{\text{exp}}}$ [78]	0.008	0.008	0.0090	0.0092	0.0093	0.0096	0.0092	0.0094	0.0095	0.0096

Note: a_{exp} is defined like $a_{\text{experience}}$

Table 8. Gap energy calculation of CsSnCl₃ and Cs₇Rb₁Sn₈Cl₂₄ perovskites in comparison with previously reported theoretical and experimental results.

Eg (eV) approximation	This work			Other works			Experiment
	GGA-PBE	GGA + U	GGA-MBJ [33]	HSEO6 [83]	PBE-SOC [83]	HSEO6-SOC [83]	
CsSnCl ₃	0.927	2.214	1.630	1.78	0.97	1.42	2.96 [81]
Cs ₇ Rb ₁ Sn ₈ Cl ₂₄	0.993	2.099					

**Figure 4.** The electronic band structure of CsSnCl₃ and Cs₇Rb₁Sn₈Cl₂₄ perovskites: (a) pure CsSnCl₃ using GGA-PBE, (b) CsSnCl₃ using GGA + U (c) Cs₇Rb₁Sn₈Cl₂₄ using GGA-PBE and (d) Cs₇Rb₁Sn₈Cl₂₄ using GGA + U.

the electron–electron exchange approximation. To address this underestimation, we employ the GGA + U approach, which allows for a partial adjustment of the band gap in comparison to the value assessed in the experiment. As shown in table 7, we varied the value of U_{eff} from 1 to 9 in order to find the best U_{eff} that aligns well with the experimentally obtained results. Our findings conclude that $U_{\text{eff}} = 9$ eV, matching the energy band of 2.214 eV (figure 4(b)), is the most suitable value, as

observed in the experimental work. Table 8 represents different approaches for predicting the band gap of CsSnCl₃ and confirms that GGA + U is the best calculation method for achieving accuracy with experimental results.

As displayed in figure 4(c), Cs₇Rb₁Sn₈Cl₂₄ is a direct band gap semiconductor. Other symmetry sites (X–M–R) also exhibit a direct band gap. Our calculations using GGA-PBE indicate that the direct band gap of Cs₇Rb₁Sn₈Cl₂₄ is

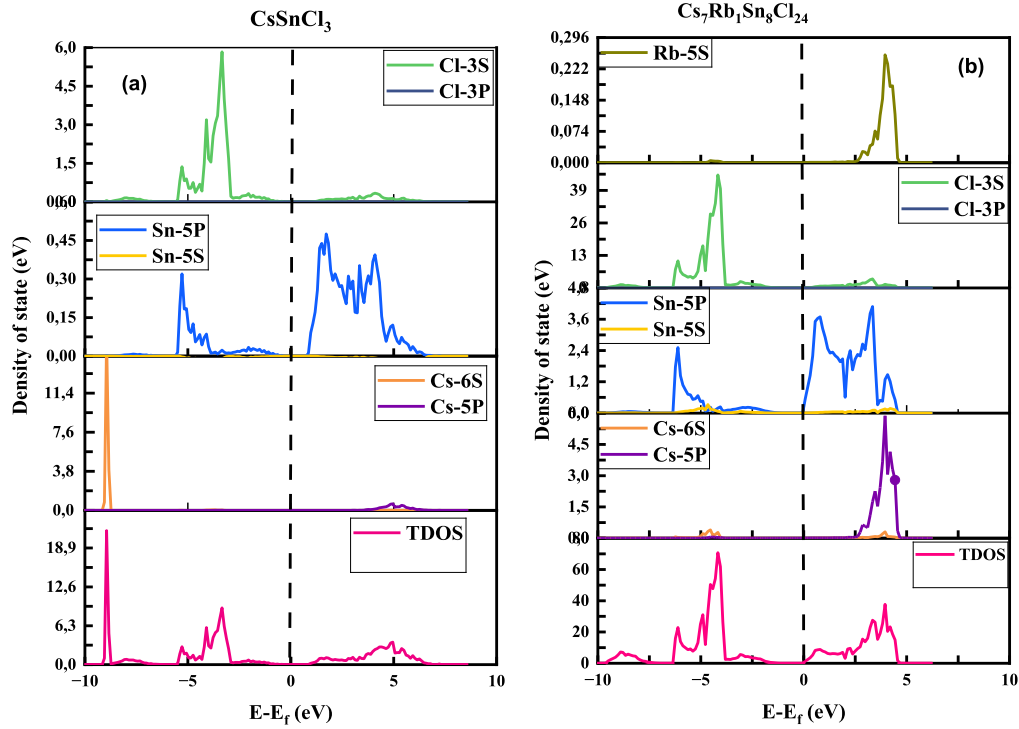


Figure 5. The electronic band structure of (a) CsSnCl_3 perovskite and (b) $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$.

0.993 eV. Furthermore, when using GGA + U approximation, we obtained a band gap of 2.099 eV (see figure 3(d)). This value of $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ was smaller behind the band gap of the pure CsSnCl_3 material.

3.4.2. Density of states (DOS). Figure 5 displays the partial (PDOS) and total doped and pure DOS of CsSnCl_3 perovskite. According to figure 5(a), the VB of CsSnCl_3 is mainly formed of a Cl-3p orbital, with notable contributions from the Cs-6s orbital and minor contributions from the Sn-5s, Sn-5p, Cs-5p, and Cl-3s orbitals. Conversely, the CB is primarily composed of a Sn-5p orbital, with a minor contributory role from the Cl-3s orbital. The $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ PDOS analysis shows that the Cl and Sn orbitals are similar to those in pure CsSnCl_3 . However, the Cs and Rb contributions are significantly pronounced in the CB, specifically Cs-5p and Rb-5s orbitals. This provides valuable information regarding the dopant states at the A-site of the perovskite-type ABX_3 that contribute to band structure changes. Figure 5 also shows that the general DOS profiles of the doped compositions are nearly identical to pure CsSnCl_3 . However, the CBs of the doped compositions are shifted slightly toward the lower energy zone. A new peak emerges in the VB near the EF in $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ due to the presence of Rb dopant states.

In order to identify the mobility of electrons and holes in the perovskite, we utilize the relationship between mobility (μ), effective mass (m^*) and electron charge (e) expressed by the formula:

$$\mu = \frac{e\tau}{m^*}. \quad (32)$$

Here, τ refers to the relaxation time. The transport properties of perovskite materials were calculated by Mahmood *et al* [59]. Following the approach recommended in the literature, a constant relaxation time of 10^{-14} s was used for the perovskite materials and maintained constant throughout the calculations in BoltzTraP code. Using formula (32), we obtained the effective mass values for both electrons and holes. The resulting value for me^* is found to be $0.300m^+$, where m^+ represents the free electron mass. Likewise, by analyzing the second derivative of the energy data near the edges of the VB, we can determine the effective mass of holes (m_h^*) using the same method. The calculated value for m_h^* is found to be $0.263m^+$. The density of states in the VB is higher than in the CB, as evident from the DOS curve. This implies that the effective mass of holes is more significant than that of electrons. The output of electron mobility (μ_e) and hole mobility (μ_h) is $58.58 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and $67.10 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, respectively, with a constant relaxation time (τ) of 10^{-14} s. The effective DOS of the VBs and CBs (N_{eff}) is computed using formula (19). The computed values of the VB (N_v) and the CB (N_c) band are 1.319×10^{20} and $5.401 \times 10^{19} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, respectively.

3.5. Optical properties

We evaluated the optical aspects of cubic CsSnCl_3 and $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ perovskite using GGA + U method. We examined specifically the absorption coefficient, reflectivity (R), and refractive index (n). The analysis included photon energies up to 20 eV to understand how the material responds to solar radiation. The absorption graph (figure 6) showed that

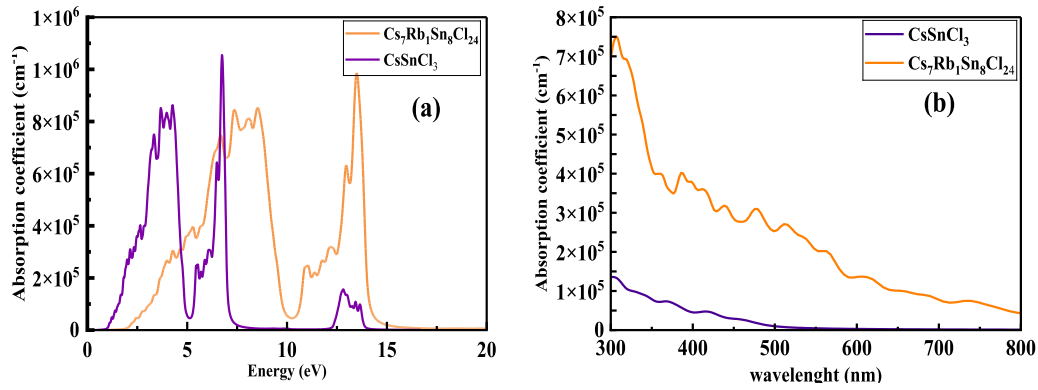


Figure 6. Optical absorption (α) of $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ using GGA + U: (a) absorption as a function of incident photon energies and (b) absorption as a function of wavelength.

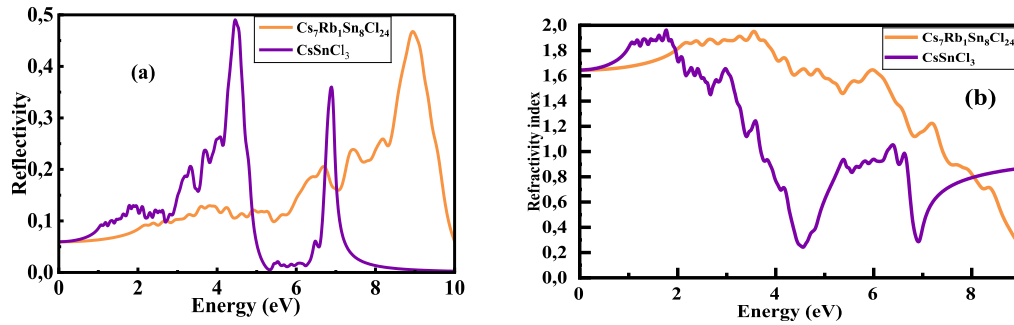


Figure 7. (a) Reflectivity and (b) refractivity of $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ using the GGA + U approximation.

$\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ exhibited a broad absorption range between 0.33 and 10 eV (visible region), with an absorption peak observed at 14.3 eV (UV). Compared to pure CsSnCl_3 , we observe that incorporation of Rb in the A-site of the perovskite shifts the absorption to the right as a function of energy. Figure 6(b) clearly shows that the absorption in the visible region of $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ is more significant than that of pure CsSnCl_3 , supporting a wider absorption in UV radiation. Adding Rb to the CsSnCl_3 structure enhanced the material's stability to absorb visible and UV radiation from the solar spectrum. By examining the plot of the absorption coefficient (α), it becomes apparent that the optical absorption phenomenon initiated at 2.099 eV. Interestingly, this specific energy point, also known as the threshold point, aligns precisely with the findings from the earlier calculated band structures. Therefore, $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ holds excellent promise as a candidate for effective solar spectrum utilization, having the potential to significantly improve photovoltaic cell efficiency. The $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ perovskite shows significantly higher absorbance than the pure CsSnCl_3 material, as reported in the literature [84]. This is due to the electrons being able to transition more efficiently from the VB to the CB under visible light exposure. This transition is supported by the formation of intermediate states.

Figure 7(a) describes the reflectivity spectra of CsSnCl_3 and Rb-doped CsSnCl_3 for light energy levels up to 10 eV. The

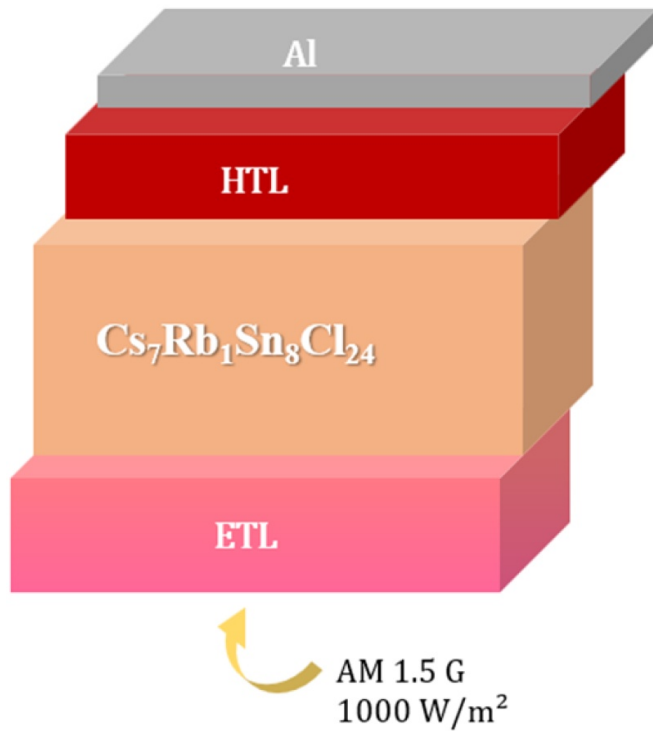
maximum reflectivity peak is observed for CsSnCl_3 at 4.33 eV, while $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ exhibits lower reflectivity, explained by a high absorption for sunlight. Figure 7(b) illustrates the refractive index of cubic CsSnCl_3 and Rb-doped CsSnCl_3 perovskite materials. The refractive index indicates how light behaves when interacting with this specific type of material. The refractive index, $n(\omega)$, indicates how transparent a material is to different spectral radiation. The refractive index is a critical parameter in the design of various optoelectronic devices, including solar cells, detectors and photonic crystals. The refractive index values were measured within the electromagnetic spectrum's visible range, ranging from 2.8 to 8.6 eV, with a notable resonance peak observed at 9.1 eV (UV), when we observe that the refractive index for both perovskites decreases sharply as photon energy increases.

3.6. $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ Sn-based solar cell

Using the parameters obtained through DFT (see table 9) in the study of $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ as the absorber layer, we investigated the functioning of an n-i-p planar PSC with a component structure of HTL/ $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ /ETL/FTO through SCAPS simulation. Figure 8 shows the simulated device architecture of the solar cell. The initial defect density in the absorber layer's bulk is assumed to be 10^{16} . The interface defects at the $\text{CuSbS}_2/\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ and $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}/\text{WS}_2$ interfaces

Table 9. Electronic parameters of the absorber layer.

Properties	Cs ₇ Rb ₁ Sn ₈ Cl ₂₄
Thickness (μm)	0.450
E_g (eV)	2.099
χ (eV)	3.930
ϵ_r	6.05
N_c (cm^{-3})	5.401×10^{19}
N_v (cm^{-3})	1.319×10^{20}
μ_n (cm^2/Vs)	58.58
μ_p (cm^2/Vs)	67.10
N_t (cm^{-3})	1×10^{15}

**Figure 8.** Device configuration of perovskite solar cell.

are considered to be single and neutral. Table 10 summarizes the defect densities within the absorber and at the interfaces.

The findings presented in table 11 show a significant enhancement in the performance of different structures, with an increase in efficiency from 12.80% to the highest recorded value of 17.39%. This suggests that the CuSbS₂/Cs₇Rb₁Sn₈Cl₂₄/WS₂/FTO configuration is the best choice for implementing solar panels. However, it is important to note that further optimization is still required for this cell.

3.6.1. Impact of thickness on the performance of the device. Figure 9 shows the thickness evolution, within the range 0.1 to 20 μm , for the absorber, CuSbS₂ and WS₂ layers and its effect on various electrical parameters, including V_{oc} , J_{sc} , FF and PCE. The results show that for CuSbS₂ (HTL) and Cs₇Rb₁Sn₈Cl₂₄ the efficiency increases with thickness up to their respective optimal values of 20 μm and 2.2 μm after a

gradual decrease is observed. Conversely, for WS₂ (ETL), the efficiency decreases immediately after reaching its optimal thickness of 0.318 μm .

The improvement in efficiency is attributed to the rise in current density as the thickness of the absorber layer is augmented. A thicker absorber layer enhances carrier generation by exposing more of the absorber material to light, generating a greater number of electron–hole pairs and increasing electron mobility, as evidenced through greater J_{sc} . V_{oc} rises as the layer thickness rises from 0.1 μm to 5 μm and stabilizes. The FF decreases as the perovskite and WS₂ layer thickness increases. This decline can be attributed to the recombination of charge carriers and the subsequent reduction in carrier lifetime in the perovskite material [85]. Furthermore, the increase in series resistance also contributes to the decrease in FF [86].

3.6.2. Impact of defect density. We aimed to explore the effect of the absorber and interface defect density on the power output of the cell, as shown in figure 10, when its range of values is balanced between 10^9 and 10^{21} cm^{-3} . The findings reveal a stable state in PCE until 10^{15} cm^{-3} , after which it drops slowly. Similarly, the graph shows the same trend for V_{oc} and J_{sc} because the defect densities in perovskite materials control to the formation energy, which is correlated to the precursor used, partial pressure and temperature conditions during synthesis. In the case of Cs₇Rb₁Sn₈Cl₂₄, the low energy of formation of about -0.7483 eV results in the creation of defects with shallow levels. This characteristic leads to a longer electron–hole diffusion length and higher V_{oc} levels. Conversely, defects possessing deep levels need a higher formation energy, leading to negative effects on both electron and hole diffusion lengths and V_{oc} levels [87].

In figures 11 and 12, we show a quasi-stable defect density for the Cs₇Rb₁Sn₈Cl₂₄ ETL interface, as well as for HTL/Cs₇Rb₁Sn₈Cl₂₄ in relation to J and V . The defect density slowly drops, eventually returning to a stable state with very low performance. The increased defect density inside the perovskite films promotes increased recombination rates, reducing both diffusion length and the lifetime of the charge carriers [88]. As the defect energy level rises, the FF progressively increases, as seen in figure 12. However, FF depends on both V_{oc} and the recombination process in the depletion region, while theoretically, FF is a function of V_{oc} .

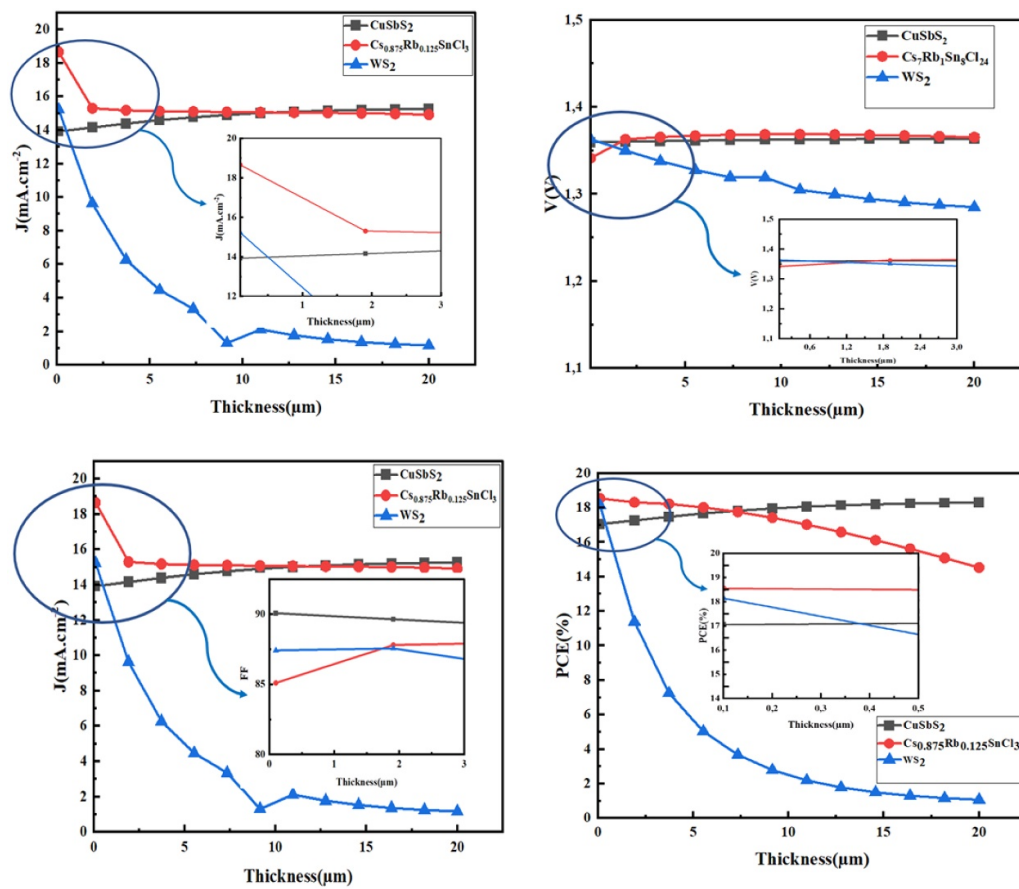
3.6.3. Effect of parasitic resistance. How series and shunt resistance affect the performance of the Cs₇Rb₁Sn₈Cl₂₄ PSC can be seen in figure 13. The shunt resistance was varied between 1000 and 10 000 $\Omega \cdot \text{cm}^2$, while the series resistance ranged from 1 to 10 $\Omega \cdot \text{cm}^2$. The graphs revealed a decrease in series resistance with increased shunt resistance. This observation aligns closely with experimental findings [89], indicating that for optimal performance a PSC needs low series resistance and high shunt. This equilibrium is essential for ensuring efficient current flow without significant losses due to internal resistances. The optimum values for series and shunt

Table 10. Defect density at the interfaces and the absorber.

	CuSbS ₂ /Cs ₇ Rb ₁ Sn ₈ Cl ₂₄	Cs ₇ Rb ₁ Sn ₈ Cl ₂₄ /WS ₂	Cs ₇ Rb ₁ Sn ₈ Cl ₂₄
Location	Interface	Interface	Bulk
Defect type	Neutral	Neutral	Neutral
Energy distribution	Single	Single	Gaussian
Energy level with respect to E_v	0.6	0.6	0.6
Total density	1×10^{13}	1×10^{13}	1×10^{15}

Table 11. Various output structures based on Cs_{0.875}Rb_{0.125}SnCl₃ computed using SCAPS.

Structure	$J_{(mA.cm^{-2})}$	$V_{(V)}$	$FF\%$	$\eta\%$
Cu ₂ O/Cs ₇ Rb ₁ Sn ₈ Cl ₂₄ /TiO ₂ /FTO	10.643	1.399	85.96	12.80
CuI/Cs ₇ Rb ₁ Sn ₈ Cl ₂₄ /WS ₂ /FTO	12.207	1.919	77.98	13.10
NiO/Cs ₇ Rb ₁ Sn ₈ Cl ₂₄ /WS ₂ /FTO	13.919	1.919	78.95	13.94
Cu ₂ O/Cs ₇ Rb ₁ Sn ₈ Cl ₂₄ /ZnO/FTO	10.116	1.56	88.96	14.06
Cu ₂ O/Cs ₇ Rb ₁ Sn ₈ Cl ₂₄ /SnS ₂ /FTO	10.384	1.561	88.85	14.40
CuO/Cs ₇ Rb ₁ Sn ₈ Cl ₂₄ /WS ₂ /FTO	10.022	1.065	68.94	14.70
CuSbS ₂ /Cs ₇ Rb ₁ Sn ₈ Cl ₂₄ /SnS ₂ /FTO	12.896	1.319	87.45	14.88
CuSbS ₂ /Cs ₇ Rb ₁ Sn ₈ Cl ₂₄ /WS ₂ /FTO	14.302	1.361	89.38	17.39

**Figure 9.** Effect of thickness on the performance of the Cs₇Rb₁Sn₈Cl₂₄ device.

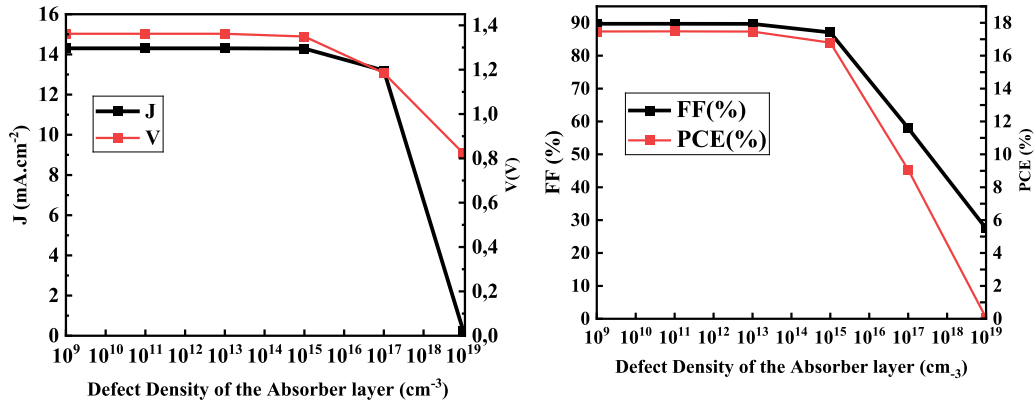


Figure 10. Influence of $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ defect density on device functionality.

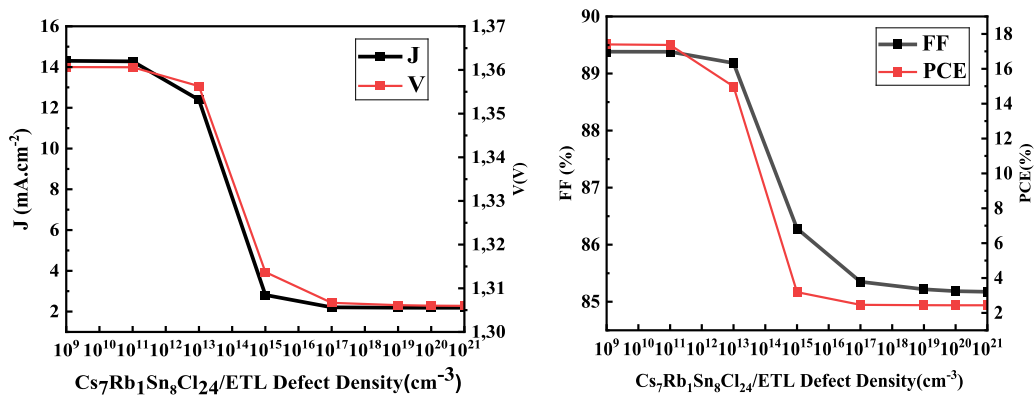


Figure 11. Impact of $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}/\text{ETL}$ interface defect density on the performance of the device.

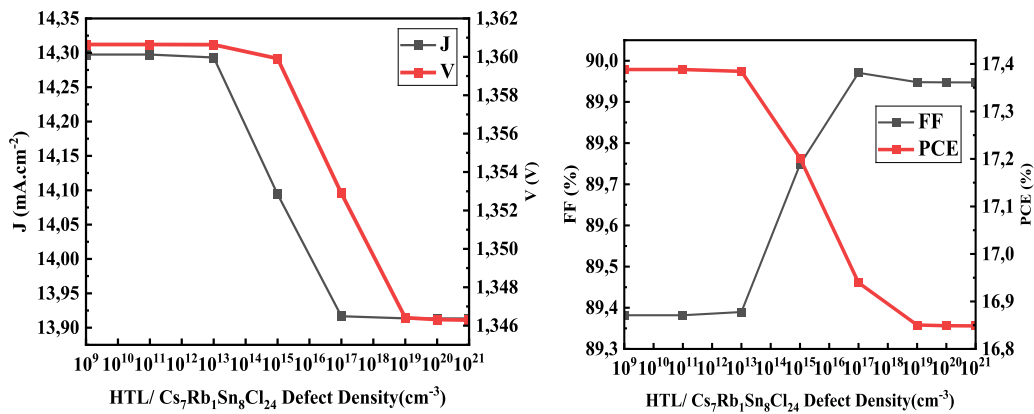


Figure 12. Impact of HTL/ $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ interface defect density on device functionality.

resistance were determined to be 1 $\Omega\cdot\text{cm}^2$ and 7000 $\Omega\cdot\text{cm}^2$, respectively.

3.6.4. Optimization of $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ PSC. Optimization of the $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ PSC within the configuration of $\text{CuSbS}_2/\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}/\text{WS}_2/\text{FTO}$ is indispensable for high-efficiency devices. Numerous key parameters have been investigated to improve performance, including absorber layer thickness and its defect density, interfacial defects between the perovskite and adjacent layers and series and shunt resistance. These optimizations collectively enhance the photovoltaic

performance and long-term stability of PSCs, facilitating their adoption in commercial settings. The final results of optimization are as follows: V_{oc} , 1.36 V; J_{sc} , 15.27 mA/cm²; FF , 88.26%; and PCE , 18.39%. To guide experimental preparation, Rb can be introduced via a mixed-cation precursor route [23], where CsCl and RbCl are combined with SnCl_2 in the desired molar ratio (e.g. $\text{Cs}_{0.875}\text{Rb}_{0.125}$, corresponding to 12.5% Rb substitution at the A-site). This is followed by spin-coating and annealing using the same protocol as for the parent CsSnCl_3 . The mixed precursor method is the most direct and reproducible for thin-film devices. It is important to note that the smaller ionic radius of Rb^+ compared to Cs^+ leads to modifications

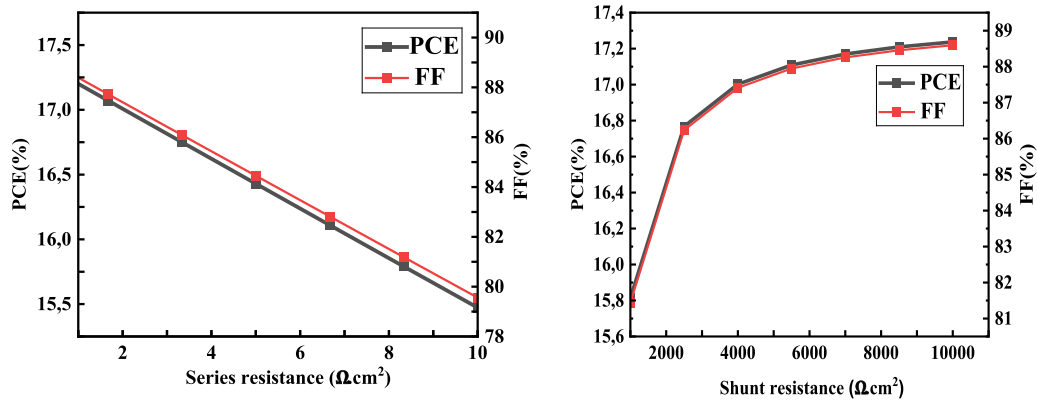


Figure 13. Influence of series and shunt resistances on device functionality.

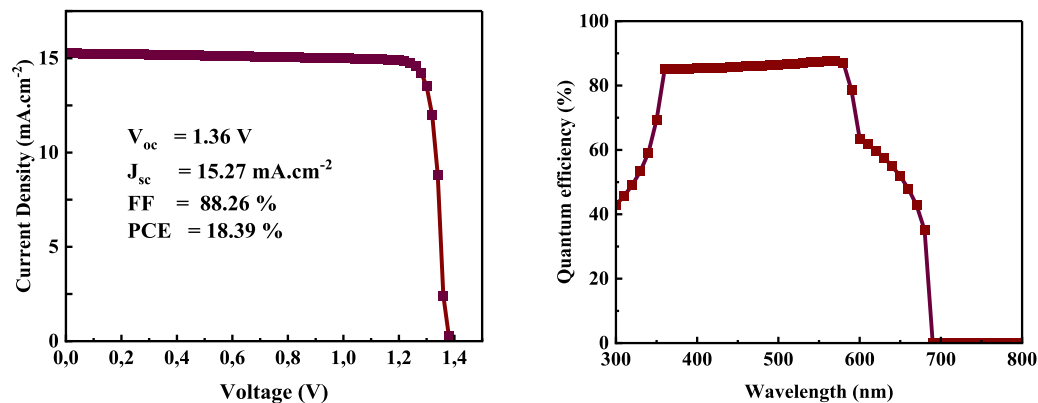


Figure 14. J - V characteristics and external quantum efficiency (EQE) of the optimal cell.

in the tolerance factor and can contract the lattice with partial substitution [76]. Therefore, we recommend conducting experimental screening within a controlled substitution range ($y = 0$ – 0.2) and performing careful structural and electronic characterization at each composition [23]. This approach balances the beneficial effects of Rb incorporation with the need to avoid excessive resistance, which can degrade the FF.

Figure 14 displays the J - V characteristics and quantum efficiency (QE) of the optimal cell with the $\text{CuSbS}_2/\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}/\text{WS}_2/\text{FTO}$ configuration. In this investigation, the QE refers explicitly to the external QE (EQE). The QE associated with different wavelengths (λ) was examined in the range 300–900 nm, as illustrated in figure 14. The J - V characteristics of PSCs are significantly impacted by QE, which directly affects the overall performance of the cells. QE measures how efficiently incident photons are converted into charge carriers (electrons and holes) in a solar cell. A high QE across the solar spectrum increases the short-circuit current density in the J - V curve, as more photons contribute to the photocurrent [90]. Increased QE reduces recombination losses and improves charge carrier extraction, resulting in an enhanced FF with steeper slopes near the open-circuit voltage and short-circuit conditions in the J - V curve. This ultimately leads to greater overall PCE. In addition, the stability and reliability of PSCs rely on maintaining high QE under various lighting conditions to ensure consistent performance over time.

Moreover, experimental studies on other hybrid/inorganic perovskites have shown that the introduction of Rb (or other small cations) can enhance film quality and photovoltaic performance parameters [91], such as a higher V_{oc} . That finding aligns significantly with our investigation.

4. Conclusion

This study has successfully explored the potential of $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ as a Pb-free perovskite material for solar cells. Utilizing DFT calculations with VASP code, a thorough analysis was conducted on this novel material's structural, elastic, mechanical, thermal and optical properties. A comprehensive examination indicates that the cubic structure of $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ displays favorable characteristics, positioning it as a promising candidate for utilization in photovoltaic applications. Our DFT calculations reveal that incorporating Rb improves both structural stability and mechanical strength of the material, making it a more ductile material for manufacturing of solar cells. The elastic properties suggest flexibility and durability, which are essential for long-term operational stability. Thermal analysis indicates appropriate thermal conductivity and stability, ensuring effective heat dissipation and sustained material integrity during operation.

The optical properties of cubic $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$, specifically, the refractive index, reflectivity, and absorption spectrum, prove the perovskite's suitable band gap and significant absorption coefficient in the visible and ultraviolet range, key for efficient light absorption. Furthermore, we examined the performance of $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ as an absorber layer in solar cells utilizing SCAPS software. The configuration used was $\text{CuSbS}_2/\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}/\text{WS}_2/\text{FTO}$. Our analysis considered various parameters such as absorber layer thickness, ETL and HTL, defect density, interfacial properties, and shunt and series resistances. Computational results suggest that optimizing these parameters can lead to a high V_{oc} of 1.36 V, J_{sc} of 15.27 $\text{mA}\cdot\text{cm}^{-2}$, FF of 88.26 and PCE of 18.39%. These findings demonstrate that $\text{Cs}_7\text{Rb}_1\text{Sn}_8\text{Cl}_{24}$ significantly improves the voltage and efficiency of Sn-based solar cells, highlighting its potential as an auspicious eco-friendly Pb-free perovskite materials for sustainable solar cell technology. Future research should focus on experimental validation of these theoretical predictions. We present this work as a foundation to guide and motivate future synthesis and device fabrication studies. Furthermore, we aim to develop a novel solution to creating solar cells for utility-scale applications that are 100% non-toxic and sustainable for both humans and the environment.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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